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Counterions between walls with surface charge modulations: Exact Poisson-Boltzmann solutions

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Abstract We study the thermal equilibrium of a classical system of identical point charges moving between two walls with parallel surfaces at a distance d , charged symmetrically with position-dependent surface charge densities. Specifically, we study the effect of surface charge modulation on the effective force (pressure) between the walls in the Poisson-Boltzmann limit. Based on Monte Carlo simulations at high temperatures, it is predicted that surface charge modulation reduces the pressure between the walls compared to uniformly charged wall surfaces with the same average surface charge density. We restrict ourselves to surface charge modulations in only one direction and, using a general solution of the two-dimensional Liouville equation, we construct exact solutions for the electrostatic potential. The surface charge density on the walls is generated inversely from this potential, which means that our exact results apply to a limited set of models with a specific variation of the surface charge density with distance d . Explicit analytical results show that surface charge modulation can both increase (small distances d) and decrease (large d) the pressure between the walls.

Keywords Coulomb fluids; counterions only; electric double layer; surface charge modulation; Poisson-Boltzmann theory; general solution of the two-dimensional Liouville equation.

1 Introduction

As a result of solvation and screening, large colloids immersed in a polar solvent such as water release mobile particles from their surfaces which are “counterions” with respect to the surface charge density created on the colloid surface. Although mobile ions in Coulomb fluids are generically of both

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signs, one can approach experimentally the limit of deionized (or salt-free) suspensions [1–3]. Counterions in the vicinity of a charged colloid represent the simplest example of an electric double layer (EDL) [4–6]. To simplify the theoretical description of the complex Coulomb problem, the curved surface of large colloids is usually replaced by a planar one, the modulated shape of the fixed charge density on the colloid surface is replaced by a uniform one, and the dielectric jump between the interior of the colloid and the solvent is ignored. In general, two geometries are studied: a single EDL and two parallel EDLs. The study of the effective interaction between two similarly charged EDLs, mediated by counterions, is of particular experimental and theoretical interest [7] due to the counterintuitive phenomenon of effective like-charge attraction observed in computer simulations [8–10] and experiments [11–15]. The relative simplicity of the studied models with uniform surface charge densities allows the development of a systematic treatment of both opposite weak-coupling [16–18] and strong-coupling [19–24] limits.

The first theoretical approaches to EDLs with heterogeneous surface charge densities were based on liquid-state approximations [12, 25, 26]. In the weak-coupling regime, analytic perturbation studies in combination with Monte Carlo simulations [27, 28] suggest an increase in the average counterion density near an inhomogeneously charged surface compared to a surface charged uniformly with the same average charge density. In the case of two parallel walls with modulated surface charge densities the increase in the counterion density at the surfaces is accompanied by a decrease in the counterion density in the midplane, which is argued to imply a decrease in the pressure between the walls [29, 30]. This phenomenon was indeed observed in a 2D exactly solvable version of parallel EDLs with modulated surface charge density at the free-fermion coupling $\Gamma = 2$ [31].

The contact value theorem for EDLs with uniform surface charge densities [32–35] has been generalized to EDLs with modulated surface charge densities in Ref. [36]. Unlike the uniform case, the pressure formula includes not only the particle density at the wall surface, but also the particle density profile between the walls; therefore, its application is limited.

In this paper, we focus on the weak coupling regime described by the Poisson-Boltzmann (PB) mean field theory, initiated long ago by Gouy [37] and Chapman [38]. Several exact solutions have been found for the counterion density profile around a single planar EDL with specific forms of surface charge modulations along only one direction [39]. Explicit results for the planar geometry confirm the increase in the density of counterions on the wall due to the modulation of the surface charge density. A similar result can be expected in the case of two symmetric parallel EDLs, leading to a decrease in the counterion density in the midplane between the walls. As we show in this work, although the PB pressure is determined by the particle density in the midplane between the walls, there is an additional (positive) contribution to the pressure due to the variation of the electrostatic potential along the midplane. This positive contribution can eliminate or even overcome the negative contribution caused by the decrease in particle density in the midplane between the walls.

This work extends previous exact solutions for a single EDL with modulated surface charge density [39] to a pair of parallel EDLs at distance d , symmetrically charged with the same surface charge densities modulated in only one direction. The method is based on the general formula for exact solutions of the 2D Liouville equation given in Ref. [40]. The task is nontrivial, since it is necessary to choose only physically acceptable “trial” functions that, when substituted into the general formula for the electrostatic potential, do not produce singularities (i.e., additional point charges) at any point in the space accessible to mobile counterions. Another requirement is that the inhomogeneous surface charge densities be the same for each of the two wall surfaces. Since the surface charge density is generated inversely from the exact electrostatic potential, changing the distance between the walls d leads to a change in the surface charge density profile. This means that our exact results apply to a limited set of models with a specific variation of the surface charge density with distance d , and not to a general model with an arbitrary fixed charge distribution. The main result of the analysis of our exactly solvable family is that surface charge modulation increases/decreases the pressure for small/large wall distances. This result corrects previous mean-field ideas about the effect of surface charge modulation on the effective wall interaction.

The paper is organized as follows. The general formalism of the mean-field PB approach for two parallel EDLs charged symmetrically with modulated surface charge density is reviewed in section 2. The standard case of uniform surface charge density is briefly summarized in section 3. The systematic expansion of the electrostatic potential and counterion density when the variation of the surface charge density around its mean value is weak is elaborated in section 4. Section 5 deals with the exact solution for the geometry of two parallel walls. Section 5.1 recapitulates the general Crowdy formula [40], which covers all possible solutions of the 2D Liouville equation. A particular solution of the 2D Liouville equation physically suitable for the electrostatic potential is discussed in section 5.2. Section 6 summarizes the necessary conditions for the regularity of the exact solution of the potential. Section 7 discusses the physical models that correspond to the specific exact solutions of the 2D Liouville equation and provides an analysis of the obtained exact results. Emphasis is put on the specific characteristics of the modulated surface charge densities generated from the exact electrostatic potential and the effect of the wall spacing and surface charge modulation on the pressure. Section 8 is a short summary with concluding remarks.

2 PB approach to two-wall geometry

2.1 General formalism for modulated surface charge density

In a three-dimensional Euclidean space of points with Cartesian coordinates $\mathbf{r} = (x, y, z)$, let there exist a pair of parallel walls surfaces perpendicular to the x axis at a distance d , the left one being located at $x = -d/2$ and the right one at $x = d/2$. The longitudinal coordinates $y \in [-L_y/2, L_y/2]$ and $z \in [-L_z/2, L_z/2]$ are unbounded, $L_y, L_z \rightarrow \infty$. Each of the walls carries the same surface charge density $\sigma(y, z)e$, where e denotes a unit charge. To

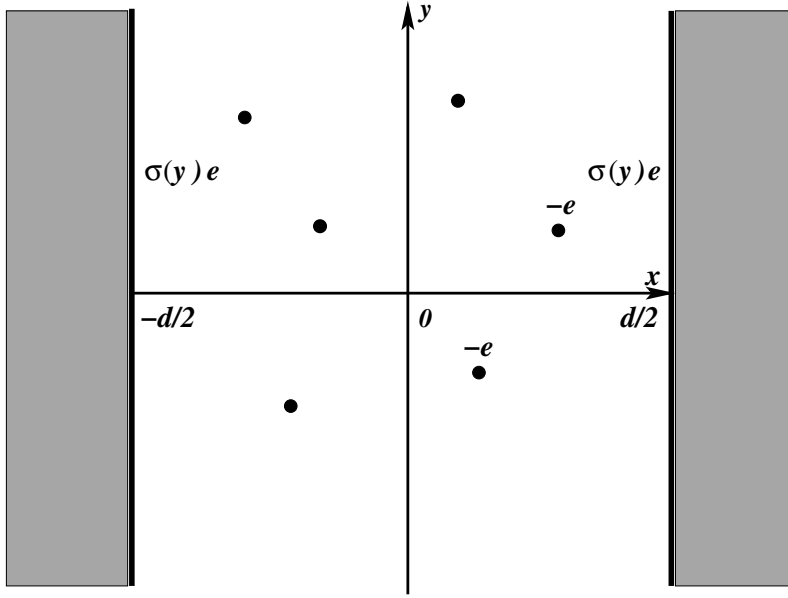


Fig. 1 Geometry of two parallel walls with surfaces at a distance d , one at $x = -d/2$ and the other at $x = d/2$. The symmetric surface charge density on the walls $\sigma(y)e$ depends on the coordinate y perpendicular to x . Point particles with charge $-e$ represented by hard disks move between the walls.

obtain exact results, we restrict ourselves to surface charge densities that vary only in one direction, say $\sigma(y, z) \equiv \sigma(y)$. Then all quantities considered in this work will depend on coordinates x and y , but not on z , i.e. the problem is effectively two-dimensional (2D). In the original formulation of the problem, when particles are released from the colloidal surfaces directly into the solvent, it holds that $\sigma(y) \geq 0$. If some particles can return to some locations on the colloidal surfaces, then this restriction is absent and $\sigma(y)$ can be negative in certain intervals of the y axis.

Identical point particles of charge $-e$, called “counterions”, move inside the region $-d/2 < x < d/2$, see Fig. 1. The particles are immersed in a medium with dielectric constant ε and interact in pairs via the Coulomb potential $1/(\varepsilon r)$ (in Gauss units). The dielectric constant of the walls is assumed for simplicity to be the same as the dielectric constant of the medium, i.e. there are no electrostatic image charges. The system of particles is in thermal equilibrium at the temperature T , or the inverse temperature $\beta = 1/(k_B T)$, where k_B is the Boltzmann constant.

Let $n(x, y)$ be the mean particle density (per unit surface of either of the walls) and $\rho(x, y) = -en(x, y)$ be the corresponding particle charge density at point $\mathbf{r}$. The condition for overall electroneutrality is

$$2 \int_{-L_y/2}^{L_y/2} dy \sigma(y)e + \int_{-d/2}^{d/2} dx \int_{-L_y/2}^{L_y/2} dy \rho(x, y) = 0. \quad (2.1)$$

The mean electrostatic potential $\psi(x, y)$ satisfies Poisson's equation [41]

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\frac{4\pi}{\varepsilon} \rho. \quad (2.2)$$

The normal derivative of ψ on the wall surface is related to the surface charge density as [41]

$$\left. \frac{\partial \psi(x, y)}{\partial x} \right|_{x=\pm d/2} = \pm \frac{4\pi \sigma(y) e}{\varepsilon}. \quad (2.3)$$

These boundary conditions are consistent with the neutrality requirement (2.1), as one can show by integrating Poisson's equation (2.2) over the entire space between the walls.

There are two relevant lengths. The distance between two unit charges e immersed in a medium with dielectric constant ε at which they interact with thermal energy $k_B T$ is called the Bjerrum length

$$\ell_B \equiv \frac{\beta e^2}{\varepsilon}. \quad (2.4)$$

The distance from a wall charged with uniform surface charge density σe at which the potential energy of a unit charge e equals the thermal energy $k_B T$ is the Gouy-Chapman length [37, 38]

$$\mu = \frac{1}{2\pi \ell_B \sigma}. \quad (2.5)$$

The coupling constant Ξ is the ratio of these two lengths:

$$\Xi = \frac{\ell_B}{\mu}. \quad (2.6)$$

The weak coupling regime $\Xi \ll 1$ is adequately described by the mean-field PB theory [37, 38]. Within the PB theory, the particle density at a point is proportional to the Boltzmann weight of the mean electrostatic potential (multiplied by the particle charge $-e$) at that point:

$$n(\mathbf{r}) = f_0 e^{\beta e \psi(\mathbf{r})}, \quad (2.7)$$

where f_0 is the normalization constant. In terms of the reduced potential $\phi = \beta e \psi$, this relation reads as

$$n(\mathbf{r}) = f_0 e^{\phi(\mathbf{r})}. \quad (2.8)$$

Let us introduce dimensionless coordinates

$$\tilde{x} = Kx, \quad \tilde{y} = Ky, \quad (2.9)$$

where

$$K = \sqrt{2\pi \ell_B f_0} \quad (2.10)$$

is a free normalization parameter for now. Poisson's equation (2.2) can be rewritten for the reduced potential as the standard 2D Liouville equation

$$\frac{\partial^2 \phi}{\partial \tilde{x}^2} + \frac{\partial^2 \phi}{\partial \tilde{y}^2} = 2e^\phi \quad (2.11)$$

and the boundary conditions (2.3) are expressed as

$$\left. \frac{\partial \phi(\tilde{x}, \tilde{y})}{\partial \tilde{x}} \right|_{\tilde{x}=\pm \tilde{d}/2} = \pm \frac{4\pi\ell_B \sigma(\tilde{y})}{K}. \quad (2.12)$$

The particle density (2.8) can be expressed using the dimensionless potential as follows

$$n(\tilde{\mathbf{r}}) = \frac{K^2}{2\pi\ell_B} e^{\phi(\tilde{\mathbf{r}})}. \quad (2.13)$$

Since for symmetrically charged walls the reduced potential exhibits the reflection symmetry,

$$\phi(\tilde{x}, \tilde{y}) = \phi(-\tilde{x}, \tilde{y}), \quad (2.14)$$

the two symmetric boundary conditions (2.12) can be replaced by the equivalent ones

$$\left. \frac{\partial \phi(\tilde{x}, \tilde{y})}{\partial \tilde{x}} \right|_{\tilde{x}=0} = 0, \quad \left. \frac{\partial \phi(\tilde{x}, \tilde{y})}{\partial \tilde{x}} \right|_{\tilde{x}=\tilde{d}/2} = \frac{4\pi\ell_B \sigma(\tilde{y})}{K}. \quad (2.15)$$

These boundary conditions formally correspond to the configuration of one wall without surface charge density localized at $\tilde{x} = 0$ and the other wall at $\tilde{x} = \tilde{d}/2$ with inhomogeneous surface charge density $\sigma(\tilde{y})$. The electroneutrality condition (2.1) has the form

$$K \int_{-\tilde{L}_y/2}^{\tilde{L}_y/2} d\tilde{y} \sigma(\tilde{y}) = \int_0^{\tilde{d}/2} d\tilde{x} \int_{-\tilde{L}_y/2}^{\tilde{L}_y/2} d\tilde{y} n(\tilde{\mathbf{r}}). \quad (2.16)$$

For a boundary with surface charge modulation, an explicit formula for the local (osmotic) pressure at a point on the boundary was derived in the PB limit in Ref. [39], see equation (2.26) there. Within the present geometry of symmetrically charged walls with boundaries at the points $\tilde{x} = \pm \tilde{d}/2$, without any dependence of the relevant quantities on the z axis, this formula simplifies to

$$\beta P(\pm \tilde{d}/2, \tilde{y}) = n(\pm \tilde{d}/2, \tilde{y}) - 2\pi\ell_B \sigma^2(\tilde{y}) + \frac{K^2}{8\pi\ell_B} \left[\frac{\partial \phi(\pm \tilde{d}/2, \tilde{y})}{\partial \tilde{y}} \right]^2. \quad (2.17)$$

The total pressure between the walls is given by averaging the local pressure over the surface of any of the walls:

$$P(\pm \tilde{d}/2) = \lim_{\tilde{L}_y \rightarrow \infty} \frac{1}{\tilde{L}_y} \int_{-\tilde{L}_y/2}^{\tilde{L}_y/2} d\tilde{y} P(\pm \tilde{d}/2, \tilde{y}). \quad (2.18)$$

A special property of PB theory is that there are infinitely many possible representations of the local pressure identified by the coordinate $\tilde{x} \in [-\tilde{d}/2, \tilde{d}/2]$:

$$\begin{aligned}\beta P(\tilde{x}, \tilde{y}) &= n(\tilde{x}, \tilde{y}) + \frac{K^2}{8\pi\ell_B} \left\{ \left[\frac{\partial\phi(\tilde{x}, \tilde{y})}{\partial\tilde{y}} \right]^2 - \left[\frac{\partial\phi(\tilde{x}, \tilde{y})}{\partial\tilde{x}} \right]^2 \right\} \\ &= \frac{K^2}{2\pi\ell_B} \left[e^\phi + \frac{1}{4} \left(\frac{\partial\phi}{\partial\tilde{y}} \right)^2 - \frac{1}{4} \left(\frac{\partial\phi}{\partial\tilde{x}} \right)^2 \right].\end{aligned}\quad (2.19)$$

The total pressure is given by

$$P(\tilde{x}) = \lim_{\tilde{L}_y \rightarrow \infty} \frac{1}{\tilde{L}_y} \int_{-\tilde{L}_y/2}^{\tilde{L}_y/2} d\tilde{y} P(\tilde{x}, \tilde{y}). \quad (2.20)$$

To prove this, we differentiate $\beta P(\tilde{x}, \tilde{y})$ with respect to $\tilde{x}$:

$$\frac{\partial\beta P}{\partial\tilde{x}} = \frac{K^2}{2\pi\ell_B} \left\{ \frac{\partial\phi}{\partial\tilde{x}} \left[e^\phi - \frac{1}{2} \left(\frac{\partial^2\phi}{\partial\tilde{x}^2} + \frac{\partial^2\phi}{\partial\tilde{y}^2} \right) \right] + \frac{1}{2} \frac{\partial}{\partial\tilde{y}} \left(\frac{\partial\phi}{\partial\tilde{x}} \frac{\partial\phi}{\partial\tilde{y}} \right) \right\}. \quad (2.21)$$

Given the 2D Poisson equation for ϕ (2.11), the expression in square brackets vanishes and we get

$$\frac{\partial\beta P}{\partial\tilde{x}} = \frac{K^2}{4\pi\ell_B} \frac{\partial}{\partial\tilde{y}} \left(\frac{\partial\phi}{\partial\tilde{x}} \frac{\partial\phi}{\partial\tilde{y}} \right). \quad (2.22)$$

Next, we integrate both sides of this equation over $\tilde{y} \in (-\tilde{L}_y/2, \tilde{L}_y/2)$. Since $P(\tilde{x}, \tilde{y})$ and its partial derivatives are continuous, we can change the order of integration and differentiation with respect to $\tilde{x}$ on the left-hand side of Eq. (2.22) to get

$$\frac{\partial}{\partial\tilde{x}} \beta P(\tilde{x}) = \frac{K^2}{4\pi\ell_B} \lim_{\tilde{L}_y \rightarrow \infty} \frac{1}{\tilde{L}_y} \left(\frac{\partial\phi}{\partial\tilde{x}} \frac{\partial\phi}{\partial\tilde{y}} \Big|_{\tilde{y}=\tilde{L}_y/2} - \frac{\partial\phi}{\partial\tilde{x}} \frac{\partial\phi}{\partial\tilde{y}} \Big|_{\tilde{y}=-\tilde{L}_y/2} \right). \quad (2.23)$$

The partial derivatives of the electric field are finite, so

$$\frac{\partial}{\partial\tilde{x}} \beta P(\tilde{x}) = 0. \quad (2.24)$$

This means that for every value of the coordinate $\tilde{x} \in [-\tilde{d}/2, \tilde{d}/2]$, the pressure $P(\tilde{x})$ is equal to the original pressure $P(\pm\tilde{d}/2)$ defined by Eqs. (2.17) and (2.18).

The midplane between the walls $\tilde{x} = 0$ is particularly interesting, because the symmetry $\phi(\tilde{x}, \tilde{y}) = \phi(-\tilde{x}, \tilde{y})$ implies that $\partial\phi/\partial\tilde{x} = 0$ at $\tilde{x} = 0$. The local pressure is then given by

$$\beta P(0, \tilde{y}) = n(0, \tilde{y}) + \frac{K^2}{8\pi\ell_B} \left[\frac{\partial\phi(0, \tilde{y})}{\partial\tilde{y}} \right]^2. \quad (2.25)$$

We see that the local pressure is determined by the counterion density in the midplane plus a positive term due to the variation of the electric potential along the midplane. This positive contribution can overcome the negative contribution due to the decrease in particle density in the midplane. This means that a decrease in the counterion density at $\tilde{x} = 0$ does not necessarily mean a decrease in pressure between the walls.

3 Uniform surface charge density

It is useful to briefly summarize the uniform case of surface charge densities on walls $\sigma(y, z) = \sigma$, see e.g. Ref. [42]. The reduced potential ϕ depends only on the x -coordinate and the 2D Liouville equation (2.11) reduces to the one-dimensional equation

$$\frac{d^2\phi}{d\tilde{x}^2} = 2e^\phi \quad (3.1)$$

where $\tilde{x} \equiv K_0 x$; the subscript “0” in K_0 means “related to the uniform case”. Multiplying this equation by $\phi' \equiv d\phi/d\tilde{x}$ and integrating over $\tilde{x}$ we get the equation

$$(\phi')^2 - 4e^\phi = \text{const.} \quad (3.2)$$

Fixing the gauge, say $\phi(0) = 0$, the explicit solution to this equation looks like

$$\phi(\tilde{x}) = -2 \ln(\cos \tilde{x}). \quad (3.3)$$

It automatically satisfies the boundary condition $\phi' = 0$ at $\tilde{x} = 0$. The second boundary condition

$$K_0 \frac{\partial \phi}{\partial \tilde{x}} \Big|_{\tilde{x}=\tilde{d}_0/2} = 4\pi\ell_B\sigma \quad (3.4)$$

implies the relation for the dimensionless distance $\tilde{d}_0 \equiv K_0 d$

$$\tilde{d}_0 \tan\left(\frac{\tilde{d}_0}{2}\right) = \bar{d}, \quad (3.5)$$

where another dimensionless distance $\bar{d} \equiv d/\mu$ is available in the sense that it does not contain any unknown parameter like K_0 . According to (2.13), the particle density is given by

$$n(\tilde{x}) = \frac{K_0^2}{2\pi\ell_B} \frac{1}{\cos^2 \tilde{x}}. \quad (3.6)$$

The effective interaction between two parallel walls at distance d is given by the pressure $P_0(d)$; the case $P_0 > 0$ corresponds to repulsion and $P_0 < 0$ to attraction of the walls. Within the two-wall formulation (2.15) and (2.16), the contact value theorem [32–35] tells us that for the present system with counterions only the pressure is determined by the particle density at the wall contacts $\tilde{x} = \pm\tilde{d}_0/2$ as

$$\beta P_0 = n(\tilde{d}_0/2) - 2\pi\ell_B\sigma^2. \quad (3.7)$$

This exact relationship is valid not only in the high-temperature limit, but also for any temperature. As shown in the previous section, there are infinitely many possible representations of the PB pressure identified by the coordinate $\tilde{x}$:

$$\beta P_0(\tilde{x}) = n(\tilde{x}) - \frac{K_0^2}{8\pi\ell_B} \left(\frac{\partial\phi}{\partial\tilde{x}} \right)^2. \quad (3.8)$$

Choosing $\tilde{x} = 0$, we get

$$\beta P_0 = n(0). \quad (3.9)$$

Using the formula for the particle density (3.6) we obtain for the dimensionless pressure

$$\bar{P}_0 \equiv \frac{\beta P_0}{2\pi\ell_B\sigma^2} = (K_0\mu)^2 = \left(\frac{\tilde{d}_0}{\bar{d}} \right)^2. \quad (3.10)$$

Since the pressure is positive, two uniformly charged walls with the same surface charge density always repel each other in the PB regime.

In the limit $\bar{d} \rightarrow 0$, it is possible to expand the left-hand side of Eq. (3.5) in Taylor powers of small $\tilde{d}_0$ to obtain a series dependence of $\tilde{d}_0$ on $\bar{d}$. Substituting this dependence into (3.10), we obtain

$$\bar{P}_0 = \frac{2}{\bar{d}} - \frac{1}{3} + \frac{2}{45}\bar{d} + O(\bar{d}^2). \quad (3.11)$$

In the limit $\bar{d} \rightarrow \infty$, Eq. (3.5) implies that $\tilde{d}_0 \rightarrow \pi$ and the large-distance asymptotic of the pressure reads as

$$\beta P_0 \underset{\bar{d} \rightarrow \infty}{\sim} \frac{\pi}{2\ell_B} \frac{1}{\bar{d}^2}. \quad (3.12)$$

Note that this universal form of pressure does not depend on the amplitude of the surface charge density σe .

4 Infinitesimal perturbation of uniform surface charge density

Let's look at the solution of the 2D Liouville equation (2.11) by perturbing slightly the uniform solution (3.3):

$$\phi(\tilde{x}, \tilde{y}) = -2 \ln(\cos \tilde{x}) + \sum_{j=1}^{\infty} \varepsilon^j f_j(\tilde{x}, \tilde{y}), \quad (4.1)$$

where ε is a positive smallness parameter and $\tilde{x}, \tilde{y}$ are dimensionless coordinates (2.9) with a normalization parameter K (2.10) depending on ε , $K \equiv K(\varepsilon)$. The smallness of $\varepsilon \ll 1$ is related to the assumption that the variation of the surface charge density around its average value is weak, $|\sigma(y) - \sigma| \ll 1$. The expansion functions exhibit the reflection symmetry of the reduced potential (2.14), i.e.,

$$f_j(\tilde{x}, \tilde{y}) = f_j(-\tilde{x}, \tilde{y}) \quad \text{for all } j = 1, 2, \dots \quad (4.2)$$

By substituting the ansatz (4.1) into the Liouville equation (2.11) and expanding all functions in powers of ϵ , the linear term in ϵ vanishes if

$$\frac{\partial^2 f_1}{\partial \tilde{x}^2} + \frac{\partial^2 f_1}{\partial \tilde{y}^2} - \frac{2}{\cos^2 \tilde{x}} f_1 = 0, \quad (4.3)$$

the quadratic term in ϵ vanishes if

$$\frac{\partial^2 f_2}{\partial \tilde{x}^2} + \frac{\partial^2 f_2}{\partial \tilde{y}^2} - \frac{2}{\cos^2 \tilde{x}} f_2 = \frac{f_1^2}{\cos^2 \tilde{x}}, \quad (4.4)$$

and so on.

We look for the leading order function of Eq. (4.3) using the separation of variables:

$$f_1(\tilde{x}, \tilde{y}) = \varphi(\tilde{x})\psi(\tilde{y}). \quad (4.5)$$

The functions φ and ψ satisfy the ordinary differential equations

$$\frac{1}{\varphi} \frac{d^2 \varphi}{d\tilde{x}^2} = b^2 + \frac{2}{\cos^2 \tilde{x}}, \quad \frac{1}{\psi} \frac{d^2 \psi}{d\tilde{y}^2} = -b^2, \quad (4.6)$$

where b is a real (say positive) parameter. The solution for ψ has the form

$$\psi = \cos(b\tilde{y}), \quad (4.7)$$

where the prefactor is set to unity for simplicity. There are two independent solutions for φ ,

$$\varphi_{\pm}(\tilde{x}) = \frac{e^{\pm b\tilde{x}}}{2} (b \pm \tan \tilde{x}), \quad (4.8)$$

and the solution possessing the reflection symmetry $\tilde{x} \rightarrow -\tilde{x}$ is given by $\varphi(\tilde{x}) = \varphi_+(\tilde{x}) + \varphi_-(\tilde{x})$. The first expansion function is then

$$f_1(\tilde{x}, \tilde{y}) = \cos(b\tilde{y}) [\tan \tilde{x} \sinh(b\tilde{x}) + b \cosh(b\tilde{x})]. \quad (4.9)$$

The reflection-symmetric solution of (4.4) for the lower-order function f_2 is the sum of the homogeneous solution f_1 (multiplied by an arbitrary real constant c) plus the inhomogeneous part:

$$f_2(\tilde{x}, \tilde{y}; c) = cf_1(\tilde{x}, \tilde{y}) + \frac{1}{4} \left\{ b^2 \sinh^2(b\tilde{x}) + \cos^2(b\tilde{y}) \left[\frac{\sinh^2(b\tilde{x})}{\cos^2 \tilde{x}} - b^2 \cosh(2b\tilde{x}) \right] \right\}. \quad (4.10)$$

Note that the inhomogeneous part of the solution cannot be written in the form of a separation of variables.

It is difficult to continue to higher order functions f_3, f_4 , etc. because the corresponding differential equations are more difficult to solve.

5 Surface charge density modulation: Exact solutions

5.1 General solution of the 2D Liouville equation

The 2D Liouville equation (2.11) is a partial differential equation of elliptic type that has many applications in physics. Several partial exact solutions of this equation have been determined in the past. The most general solution [40] is expressible in terms of complex variables

$$z = \tilde{x} + i\tilde{y}, \quad \bar{z} = \tilde{x} - i\tilde{y} \quad (5.1)$$

as follows

$$\begin{aligned} \phi(\tilde{x}, \tilde{y}) = & -2 \ln [c_1 Y_1(z) \bar{Y}_1(\bar{z}) + c_2 Y_2(z) \bar{Y}_2(\bar{z}) \\ & + c_3 Y_1(z) \bar{Y}_2(\bar{z}) + \bar{c}_3 \bar{Y}_1(\bar{z}) Y_2(z)] + \ln [W(z) \bar{W}(\bar{z})], \end{aligned} \quad (5.2)$$

where the two independent “trial” functions $Y_1(z)$ and $Y_2(z)$ are analytic in z and exhibit nonzero Wronskian

$$W(z) \equiv Y_1(z) Y_2'(z) - Y_1'(z) Y_2(z), \quad (5.3)$$

c_1, c_2 are real constants and c_3 a complex constant, constrained by

$$|c_3|^2 - c_1 c_2 = \frac{1}{4}. \quad (5.4)$$

Here the conjugate function $\bar{f}(\bar{z})$ is defined as $\bar{f}(\bar{z}) \equiv \overline{f(z)}$.

5.2 Particular solutions for the two-wall geometry

It is instructive to reproduce the exact result for two uniformly charged walls at a distance d (3.3) with respect to the general 2D solution (5.2)-(5.4). Let us choose the following trial functions

$$Y_1(z) = e^{iz/2}, \quad Y_2(z) = e^{-iz/2} \quad (5.5)$$

and fix the constants as follows

$$c_1 = c_2 = 0, \quad c_3 = \frac{1}{2}. \quad (5.6)$$

Since

$$\frac{1}{2} [Y_1(z) \bar{Y}_2(\bar{z}) + \bar{Y}_1(\bar{z}) Y_2(z)] = \cos(\tilde{x}) \quad (5.7)$$

and $W(z) = -i$, $\bar{W}(\bar{z}) = i$, the solution (5.2) coincides with the uniform one (3.3), as intended.

Now let the trial functions be chosen as

$$Y_1(z) = e^{iz/2} [1 + ia \sinh(bz)], \quad Y_2(z) = e^{-iz/2} [1 - ia \sinh(bz)], \quad (5.8)$$

where a, b are positive real parameters and the constants c_1, c_2, c_3 are chosen as before in Eq. (5.6). These trial functions reduce themselves to the previous ones when $a = 0$. After some simple algebra, the explicit solution for ϕ is

$$\phi(\tilde{x}, \tilde{y}) = -2 \ln \cos \tilde{x} - 2 \ln g(\tilde{x}, \tilde{y}) + \ln h(\tilde{x}, \tilde{y}), \quad (5.9)$$

where

$$g(\tilde{x}, \tilde{y}) = 1 - a^2 \cosh^2(b\tilde{x}) + a^2 \cos^2(b\tilde{y}) - 2a \cos(b\tilde{y}) \tan \tilde{x} \sinh(b\tilde{x}) \quad (5.10)$$

and

$$\begin{aligned} h(\tilde{x}, \tilde{y}) &= [1 + 2ab \cosh(bz) + a^2 \sinh^2(bz)] \\ &\quad \times [1 + 2ab \cosh(b\bar{z}) + a^2 \sinh^2(b\bar{z})] \\ &= \left\{ 1 + 2ab \cosh(b\tilde{x}) \cos(b\tilde{y}) + 2a^2 \cosh^2(b\tilde{x}) \cos^2(b\tilde{y}) \right. \\ &\quad \left. - a^2 [\cosh^2(b\tilde{x}) + \cos^2(b\tilde{y})] \right\}^2 \\ &\quad + 4a^2 \sinh^2(b\tilde{x}) \sin^2(b\tilde{y}) [b + a \cosh(b\tilde{x}) \cos(b\tilde{y})]^2. \end{aligned} \quad (5.11)$$

Note that the electrostatic potential ϕ has the required reflection symmetry $\phi(\tilde{x}, \tilde{y}) = \phi(-\tilde{x}, \tilde{y})$. It is easy to verify, e.g. using the symbolic calculation program *Mathematica*, that this solution indeed satisfies the 2D Liouville equation (2.11).

The Taylor expansion of the exact ϕ (5.9) in powers of the parameter a leads to

$$\phi(\tilde{x}, \tilde{y}) = -2 \ln \cos \tilde{x} + 4af_1(\tilde{x}, \tilde{y}) + (4a)^2 f_2(\tilde{x}, \tilde{y}; c=0) + \mathcal{O}(a^3), \quad (5.12)$$

where f_1 and f_2 are the expansion functions (4.9) and (4.10) of the small- ε expansion of the electrostatic potential (4.1) obtained in the previous section. We see that a is related to ε by $a = \varepsilon/4$. The exact solution (5.9) – (5.11) explicitly includes all higher-order f -functions in the ansatz (4.1), which are difficult to obtain otherwise.

6 Conditions for regular potential solutions

The solution for the reduced electrostatic potential (5.9) must be well-defined, in particular real and finite (without any singularity), over the entire interval of possible values of x and over one period along the $\tilde{y}$ axis, i.e.

$$\tilde{x} \in [0, \tilde{d}/2], \quad \tilde{y} \in [0, 2\pi/b). \quad (6.1)$$

This puts some constraints on the values of the model parameters a, b and $\tilde{d}$. Numerical verification of the regularity of the potential (5.9) – (5.11) is laborious, because it must be performed on a sufficiently dense network of discrete points covering the entire region (6.1). Therefore, we derive all necessary and sufficient conditions for regularity analytically.

Since $\cos \tilde{x}$ under the logarithm must be positive for any value $\tilde{x}$, the positive value of $\tilde{d}$ is limited as follows

$$0 \leq \tilde{d} < \pi. \quad (6.2)$$

Further restrictions concern the functions g (5.10) and h (5.11), which are under logarithm and therefore must be positive in the entire region (6.1). Next, we present a general analysis of all necessary and sufficient conditions for the parameters a , b and $\tilde{d}$ that guarantee the positivity of the functions g and h , which is quite complex and is presented in the following text.

6.1 $g(\tilde{x}, \tilde{y}) > 0$

Let $(\tilde{x}_{\min}, \tilde{y}_{\min})$ denote the minimum of g given by (5.10) in the domain (6.1). The required positivity of $g(\tilde{x}, \tilde{y})$ is equivalent to the condition

$$g(\tilde{x}_{\min}, \tilde{y}_{\min}) > 0. \quad (6.3)$$

For a fixed value of $\tilde{x}$, the minimum of the function $g(\tilde{x}, \tilde{y})$ on the $\tilde{y}$ axis is determined by the conditions

$$\left. \frac{\partial g(\tilde{x}, \tilde{y})}{\partial \tilde{y}} \right|_{\tilde{y}=\tilde{y}_{\min}} = 0, \quad \left. \frac{\partial^2 g(\tilde{x}, \tilde{y})}{\partial \tilde{y}^2} \right|_{\tilde{y}=\tilde{y}_{\min}} > 0. \quad (6.4)$$

Since

$$\frac{\partial g(\tilde{x}, \tilde{y})}{\partial \tilde{y}} = 2ab \sin(b\tilde{y}) [\sinh(b\tilde{x}) \tan \tilde{x} - a \cos(b\tilde{y})], \quad (6.5)$$

there are three candidates for the minimum point:

$$\tilde{y}_1 = 0, \quad \tilde{y}_2 = \frac{\pi}{b}, \quad \cos(b\tilde{y}_3) = \frac{\sinh(b\tilde{x}) \tan \tilde{x}}{a}. \quad (6.6)$$

The relation

$$\left. \frac{\partial^2 g(\tilde{x}, \tilde{y})}{\partial \tilde{y}^2} \right|_{\tilde{y}=\tilde{y}_1} = 2ab^2 [\sinh(b\tilde{x}) \tan \tilde{x} - a] \quad (6.7)$$

tells us that $\tilde{y}_1$ is a minimum point if

$$a < \sinh(b\tilde{x}) \tan \tilde{x}. \quad (6.8)$$

The relation

$$\left. \frac{\partial^2 g(\tilde{x}, \tilde{y})}{\partial \tilde{y}^2} \right|_{\tilde{y}=\tilde{y}_2} = -2ab^2 [\sinh(b\tilde{x}) \tan \tilde{x} + a] \quad (6.9)$$

tells us that $\tilde{y}_2$ is always a maximum point. Finally, the relation

$$\left. \frac{\partial^2 g(\tilde{x}, \tilde{y})}{\partial \tilde{y}^2} \right|_{\tilde{y}=\tilde{y}_3} = 2b^2 [a - \sinh(b\tilde{x}) \tan \tilde{x}] [\sinh(b\tilde{x}) \tan \tilde{x} + a] \quad (6.10)$$

tells us that $\tilde{y}_3$ is a minimum point if

$$a > \sinh(b\tilde{x}) \tan \tilde{x}. \quad (6.11)$$

Let's start with $\tilde{y}_1 = 0$ which is the minimum point if $a < \sinh(b\tilde{x}) \tan \tilde{x}$. The corresponding function

$$g(\tilde{x}, 0) = 1 - a^2 \sinh^2(b\tilde{x}) - 2a \sinh(b\tilde{x}) \tan \tilde{x} \quad (6.12)$$

has a negative derivative with respect to $\tilde{x}$:

$$\begin{aligned} \frac{dg(\tilde{x}, 0)}{d\tilde{x}} &= -2a^2 b \sinh(b\tilde{x}) \cosh(b\tilde{x}) \\ &\quad - 2a \left[\frac{\sinh(b\tilde{x})}{\cos^2 \tilde{x}} + b \cosh(b\tilde{x}) \tan \tilde{x} \right] < 0. \end{aligned} \quad (6.13)$$

This means that the minimum of $g(\tilde{x}, 0)$ is reached at the endpoint $\tilde{x} = \tilde{d}/2$, where the condition (6.8) takes the form

$$a < \sinh(b\tilde{d}/2) \tan(\tilde{d}/2). \quad (6.14)$$

The positivity requirement $g(\tilde{d}/2, 0) > 0$ is equivalent to the inequality

$$1 - a^2 \sinh^2(b\tilde{d}/2) - 2a \sinh(b\tilde{d}/2) \tan(\tilde{d}/2) > 0. \quad (6.15)$$

From this inequality, it follows that

$$0 \leq a < \frac{1}{\sinh(b\tilde{d}/2)} \frac{1 - \sin(\tilde{d}/2)}{\cos(\tilde{d}/2)}. \quad (6.16)$$

Given conditions (6.14) and (6.16), the parameter a is limited by the relation

$$0 \leq a < \min \left\{ \sinh(b\tilde{d}/2) \tan(\tilde{d}/2), \frac{1}{\sinh(b\tilde{d}/2)} \frac{1 - \sin(\tilde{d}/2)}{\cos(\tilde{d}/2)} \right\}. \quad (6.17)$$

Let's continue with $\tilde{y}_3$ given by (6.6) which is the minimum point if $a > \sinh(b\tilde{x}) \tan \tilde{x}$. We need to find the minimum of the function

$$g(\tilde{x}, y_3) = 1 - a^2 \cosh^2(b\tilde{x}) - \sinh^2(b\tilde{x}) \tan^2 \tilde{x} \quad (6.18)$$

on the interval $\tilde{x} \in [0, \tilde{d}/2]$. Since

$$\begin{aligned} \frac{dg(\tilde{x}, y_3)}{d\tilde{x}} &= -2a^2 b \cosh(b\tilde{x}) \sinh(b\tilde{x}) \\ &\quad - 2 \tan \tilde{x} \sinh(b\tilde{x}) \left[\frac{\sinh(b\tilde{x})}{\cos^2 \tilde{x}} + b \tan \tilde{x} \cosh(b\tilde{x}) \right] < 0, \end{aligned} \quad (6.19)$$

the minimum of $g(\tilde{x}, y_3)$ is reached at the endpoint $\tilde{x} = \tilde{d}/2$ where the condition (6.11) reads as

$$a > \sinh(b\tilde{d}/2) \tan(\tilde{d}/2). \quad (6.20)$$

and the condition (6.3) gives

$$1 - a^2 \cosh^2(b\tilde{d}/2) - \tan^2(\tilde{d}/2) \sinh^2(b\tilde{d}/2) > 0. \quad (6.21)$$

This inequality means that

$$a < \frac{\sqrt{1 - \sinh^2(b\tilde{d}/2) \tan^2(\tilde{d}/2)}}{\cosh(b\tilde{d}/2)}. \quad (6.22)$$

Finally, the parameter a is now constrained by the relation

$$\sinh(b\tilde{d}/2) \tan(\tilde{d}/2) \leq a < \frac{\sqrt{1 - \sinh^2(b\tilde{d}/2) \tan^2(\tilde{d}/2)}}{\cosh(b\tilde{d}/2)}. \quad (6.23)$$

In order to have a non-empty set of a -values determined by (6.23), it is necessary that the lower bound is smaller than the upper bound, i.e.

$$\sinh(b\tilde{d}/2) \tan(\tilde{d}/2) < \frac{\sqrt{1 - \sinh^2(b\tilde{d}/2) \tan^2(\tilde{d}/2)}}{\cosh(b\tilde{d}/2)}. \quad (6.24)$$

This inequality is equivalent to the equation

$$\tan^2(\tilde{d}/2) \sinh^4(b\tilde{d}/2) + 2 \tan^2(\tilde{d}/2) \sinh^2(b\tilde{d}/2) - 1 < 0, \quad (6.25)$$

which leads to

$$\sinh(b\tilde{d}/2) \tan(\tilde{d}/2) < \frac{1}{\sinh(b\tilde{d}/2)} \frac{1 - \sin(\tilde{d}/2)}{\cos(\tilde{d}/2)}. \quad (6.26)$$

Combining the relations (6.17) and (6.23), we conclude that if the inequality (6.26) holds, the values of a are restricted to

$$0 \leq a < \frac{\sqrt{1 - \sinh^2(b\tilde{d}/2) \tan^2(\tilde{d}/2)}}{\cosh(b\tilde{d}/2)}. \quad (6.27)$$

Now consider the inequality opposite to that in (6.24), i.e.

$$\sinh(b\tilde{d}/2) \tan(\tilde{d}/2) > \frac{\sqrt{1 - \sinh^2(b\tilde{d}/2) \tan^2(\tilde{d}/2)}}{\cosh(b\tilde{d}/2)}. \quad (6.28)$$

This inequality can be shown to be equivalent to the one

$$\sinh(b\tilde{d}/2) \tan(\tilde{d}/2) > \frac{1}{\sinh(b\tilde{d}/2)} \frac{1 - \sin(\tilde{d}/2)}{\cos(\tilde{d}/2)}. \quad (6.29)$$

Then the restriction (6.23) does not provide any solution for a and by the restriction (6.17) the values of a are limited to the interval

$$0 \leq a < \frac{1}{\sinh(b\tilde{d}/2)} \frac{1 - \sin(\tilde{d}/2)}{\cos(\tilde{d}/2)}. \quad (6.30)$$

6.2 $h(\tilde{x}, \tilde{y}) > 0$

The function $h(\tilde{x}, \tilde{y})$ (5.11) is the sum of two squares and as such can be either positive or equal to 0. The “dangerous” equality $h(\tilde{x}, \tilde{y}) = 0$ is satisfied if the two equations

$$\sinh(b\tilde{x}) \sin(b\tilde{y}) [b + a \cosh(b\tilde{x}) \cos(b\tilde{y})] = 0, \quad (6.31)$$

$$1 + 2ab \cosh(b\tilde{x}) \cos(b\tilde{y}) + 2a^2 \cosh^2(b\tilde{x}) \cos^2(b\tilde{y}) - a^2 [\cosh^2(b\tilde{x}) + \cos^2(b\tilde{y})] = 0, \quad (6.32)$$

are valid simultaneously. Eq. (6.31) is satisfied for $\tilde{x} = 0$, or $\tilde{y} = 0$, or $\tilde{y} = \pi/b$, or $b + a \cosh(b\tilde{x}) \cos(b\tilde{y}) = 0$.

• $\tilde{x} = 0$: In this case the second equation (6.32) reads

$$a^2 \cos^2(b\tilde{y}) + 2ab \cos(b\tilde{y}) + (1 - a^2) = 0. \quad (6.33)$$

Its solution is

$$\cos(b\tilde{y}) = -\frac{b}{a} \pm \frac{1}{a} \sqrt{a^2 + b^2 - 1}. \quad (6.34)$$

There is no real solution for $\cos(b\tilde{y})$ provided that

$$a^2 + b^2 < 1. \quad (6.35)$$

Since $-1 \leq \cos(b\tilde{y}) \leq 1$, a real solution for $\tilde{y}$ is missing also if

$$a^2 + b^2 = 1, \quad b > a. \quad (6.36)$$

When $a^2 + b^2 > 1$, a real solution for $\cos(b\tilde{y})$ (6.34) must lie outside the interval $[-1, 1]$ to ensure the positivity of h . In particular, if we put a $(-)$ sign in front of the square root on the right-hand side of (6.34), we have

$$-\frac{b}{a} - \frac{1}{a} \sqrt{a^2 + b^2 - 1} < -1, \quad (6.37)$$

or equivalently,

$$a - b < \sqrt{a^2 + b^2 - 1}. \quad (6.38)$$

This inequality is definitely true if $a < b$. When $a > b$, bringing each side of the inequality to the power of 2 gives us $2ab > 1$. If we take the sign $(+)$ before the square root on the right-hand side of (6.34), we must distinguish between two cases: $a^2 > 1$, when the inequality

$$-\frac{b}{a} + \frac{1}{a} \sqrt{a^2 + b^2 - 1} > 1 \quad (6.39)$$

is equivalent to the requirement $2ab < -1$, which has no solution for positive parameters a and b , and $a^2 < 1$, when the inequality

$$-\frac{b}{a} + \frac{1}{a} \sqrt{a^2 + b^2 - 1} < -1 \quad (6.40)$$

has a solution if $a < b$ and $2ab < 1$. To conclude this part, $h > 0$ for $\tilde{x} = 0$ if one of the conditions (6.35) or (6.36) holds, or if the conditions

$$a^2 + b^2 > 1, \quad a^2 < 1, \quad a < b, \quad 2ab < 1 \quad (6.41)$$

hold simultaneously.

- $\tilde{y} = 0$: In this case the second equation (6.32) has the form

$$a^2 \sinh^2(b\tilde{x}) + 2ab \cosh(b\tilde{x}) + 1 = 0. \quad (6.42)$$

The function on the left-hand side is always positive for $\tilde{x} \in [0, \tilde{d}/2]$. Consequently, h is always positive for $\tilde{y} = 0$, which does not impose any restrictions on the possible values of the parameters a , b and $\tilde{d}$.

- $\tilde{y} = \pi/b$: In this case, the second equation (6.32) reads

$$a^2 \cosh^2(b\tilde{x}) - 2ab \cosh(b\tilde{x}) + (1 - a^2) = 0. \quad (6.43)$$

Its solution is

$$\cosh(b\tilde{x}) = \frac{b}{a} \pm \frac{1}{a} \sqrt{a^2 + b^2 - 1}. \quad (6.44)$$

There is no real solution for $\cosh(b\tilde{x})$ assuming that

$$a^2 + b^2 < 1. \quad (6.45)$$

Eq. (6.44) has no solution for $\tilde{x}$ if $a^2 + b^2 = 1$ and at the same time $b < a$. If $a^2 + b^2 > 1$, Eq. (6.44) has no solution for $\tilde{x}$ in two cases:

$$\frac{b}{a} + \frac{1}{a} \sqrt{a^2 + b^2 - 1} < 1, \quad \frac{b}{a} - \frac{1}{a} \sqrt{a^2 + b^2 - 1} > \cosh(b\tilde{d}/2). \quad (6.46)$$

- $b + a \cosh(b\tilde{x}) \cos(b\tilde{y}) = 0$: This condition is equivalent to the condition

$$-\cos(b\tilde{y}) = \frac{b}{a \cosh(b\tilde{x})}. \quad (6.47)$$

The function $1/\cosh(b\tilde{x})$ decreases with increasing $\tilde{x}$, so if

$$a < \frac{b}{\cosh(b\tilde{d}/2)} \quad (6.48)$$

there is no solution for $-\cos(b\tilde{y})$ in the interval $[0, 1]$. On the other hand, if

$$a > \frac{b}{\cosh(b\tilde{d}/2)}, \quad (6.49)$$

substituting $\cos(b\tilde{y})$ from (6.47) into the second equation (6.32) yields

$$\cosh(b\tilde{x}) = \frac{\sqrt{1 + 2ab} + \sqrt{1 - 2ab}}{2a}, \quad -\cos(b\tilde{y}) = \frac{\sqrt{1 + 2ab} - \sqrt{1 - 2ab}}{2a}. \quad (6.50)$$

Real solutions for $\tilde{x}$ and $\tilde{y}$ do not exist if

$$2ab > 1. \quad (6.51)$$

On the other hand, if

$$2ab < 1, \quad (6.52)$$

we look for values of parameters a and b such that either $\cosh(b\tilde{x})$ given by (6.50) lies outside the interval

$$1 \leq \cosh(b\tilde{x}) \leq \cosh(b\tilde{d}/2), \quad (6.53)$$

what happens if

$$0 \leq a < \frac{1}{\cosh(b\tilde{d}/2)} \sqrt{1 - \frac{b^2}{\cosh^2(b\tilde{d}/2)}}, \quad (6.54)$$

or $-\cos(b\tilde{y})$ given by (6.50) lies outside the interval

$$0 \leq -\cos(b\tilde{y}) \leq 1, \quad (6.55)$$

what happens if $a^2 + b^2 > 1$.

7 Physical model generated from potential

7.1 Derivation of physical quantities

The electric potential defined by equations (5.9), (5.10) and (5.11) is periodic along the y axis with a period equal to $2\pi/b$. According to the relation (2.15), the surface charge density on the wall surface is given by

$$\sigma(\tilde{y}) = \frac{K}{4\pi\ell_B} \frac{\partial\phi(\tilde{x}, \tilde{y})}{\partial\tilde{x}} \Big|_{\tilde{x}=\tilde{d}/2}. \quad (7.1)$$

It is also periodic along the y axis with period $2\pi/b$. The mean value of the surface charge density is then given by

$$\langle\sigma\rangle = \frac{K}{4\pi\ell_B} \frac{b}{2\pi} \int_0^{2\pi/b} d\tilde{y} \frac{\partial\phi(\tilde{x}, \tilde{y})}{\partial\tilde{x}} \Big|_{\tilde{x}=\tilde{d}/2}. \quad (7.2)$$

The particle density (2.13) and local pressure (2.25) are also periodic along the y axis with period $2\pi/b$, so the mean pressure acting on either of the walls is given by

$$\beta\langle P\rangle = \frac{K^2}{2\pi\ell_B} \frac{b}{2\pi} \int_0^{2\pi/b} d\tilde{y} \left\{ e^{\phi(0, \tilde{y})} + \frac{1}{4} \left[\frac{\partial\phi(0, \tilde{y})}{\partial\tilde{y}} \right]^2 \right\}. \quad (7.3)$$

The dimensionless pressure has the form

$$\bar{P} \equiv \frac{\beta\langle P\rangle}{2\pi\ell_B\langle\sigma\rangle^2} = \frac{8\pi}{b} \frac{\int_0^{2\pi/b} d\tilde{y} \left\{ e^{\phi(0, \tilde{y})} + \frac{1}{4} \left[\frac{\partial\phi(0, \tilde{y})}{\partial\tilde{y}} \right]^2 \right\}}{\left[\int_0^{2\pi/b} d\tilde{y} \frac{\partial\phi(\tilde{x}, \tilde{y})}{\partial\tilde{x}} \Big|_{\tilde{x}=\tilde{d}/2} \right]^2}. \quad (7.4)$$

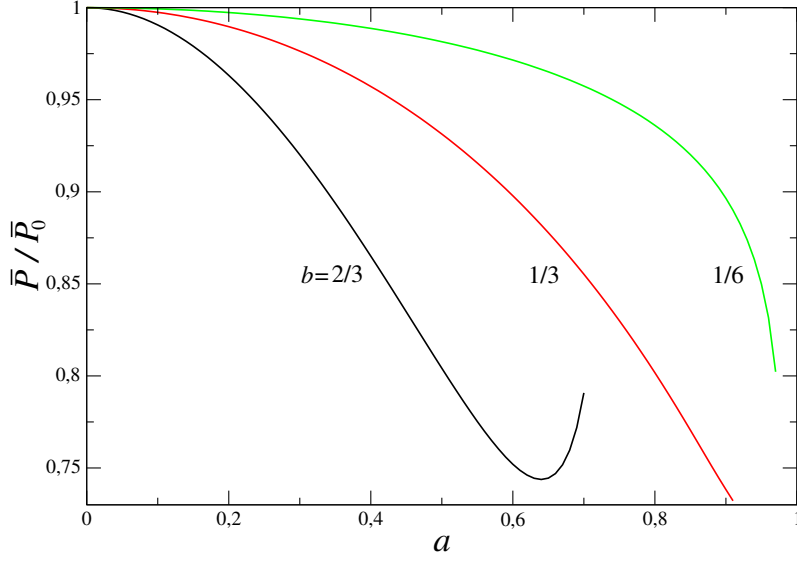


Fig. 2 Dependence of the ratio $\bar{P}/\bar{P}_0$ on the amplitude parameter a for the dimensionless distance between the walls $\tilde{d} = \pi/2$ and the values of the period parameter $b = 2/3, 1/3, 1/6$.

The relationship between the dimensionless distances $\tilde{d}$ and $\bar{d}$ follows from the definition of the latter:

$$\bar{d} \equiv 2\pi\ell_B\langle\sigma\rangle d = \tilde{d} \frac{b}{4\pi} \int_0^{2\pi/b} d\tilde{y} \frac{\partial\phi(\tilde{x}, \tilde{y})}{\partial\tilde{x}} \Big|_{\tilde{x}=\tilde{d}/2}. \quad (7.5)$$

For given values of the parameters a , b and $\tilde{d}$, this $\bar{d}$ is used to calculate the relevant quantities for the case of uniform surface charge densities, namely $\tilde{d}_0$ given by the relation (3.5) and subsequently $\bar{P}_0$ given by the relation (3.10).

7.2 Results

The dimensionless pressure and modulated surface charge density of our exactly solvable model depend on the dimensionless distance between the walls $\tilde{d}$ constrained by (6.2), the amplitude parameter a , and the period parameter b .

First, let us consider a relatively large distance $\tilde{d} = \pi/2$. The results for the ratio of the dimensionless pressure $\bar{P}$ for the model with surface charge modulation and the pressure $\bar{P}_0$ for the model with a uniform surface charge density equal to the mean value of the modulated surface charge density $\langle\sigma\rangle$ given by the relation (7.2) (see section 3) are shown in Fig. 2. Specifically, for three values of the period parameter $b = 2/3, 1/3$ and $1/6$, the dependence of the ratio $\bar{P}/\bar{P}_0$ on the amplitude parameter a is shown by solid lines. The intervals of allowed values of a are determined by the conditions for a regular

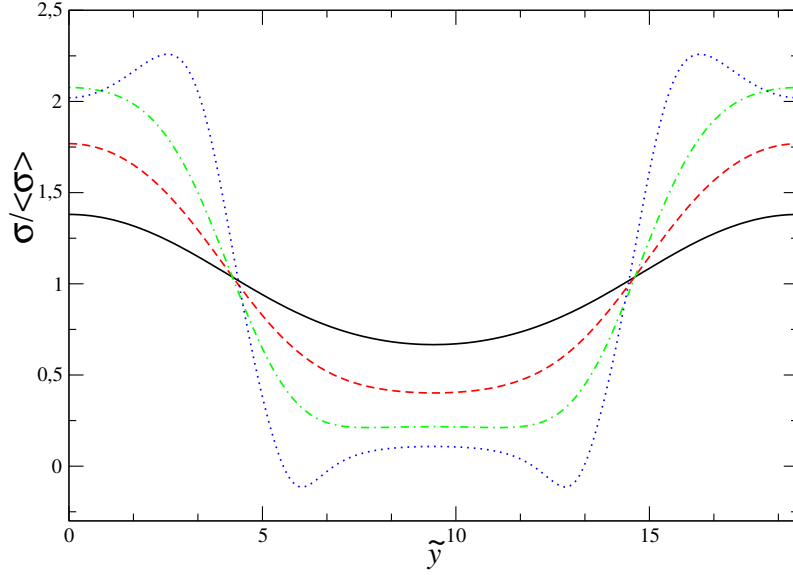


Fig. 3 Plots of the normalized modulated surface charge density $\sigma/\langle\sigma\rangle$ on the dimensionless coordinate $\tilde{y}$ for the dimensionless distance between the walls $\tilde{d} = \pi/2$ and the period parameter $b = 1/3$, plotted within one period $2\pi/b = 6\pi$. The solid line corresponds to the amplitude parameter $a = 1/5$, the dashed line to $a = 2/5$, the dash-dotted line to $a = 3/5$, and the dotted line to $a = 4/5$.

electric potential derived in section 6. It can be seen that for each value of b the ratio $\bar{P}/\bar{P}_0$ is less than 1, which means that the modulation of the surface charge density weakens the pressure between the walls. Interestingly, for $b = 2/3$ the plot of $\bar{P}/\bar{P}_0$ as a function of a is not monotonic.

The plots of the normalized modulated surface charge density $\sigma/\langle\sigma\rangle$ on the dimensionless coordinate $\tilde{y}$ for the dimensionless wall distance $\tilde{d} = \pi/2$ and the period parameter $b = 1/3$ are shown for one period $2\pi/b = 6\pi$ in Fig. 3. The plots become more pronounced with increasing value of a , as expected. The solid line ($a = 1/5$) corresponds to a slowly varying surface charge density, the change is more pronounced for $a = 2/5$ (dashed line) and $a = 3/5$ (dash-dotted line), but the surface charge is always positive with a minimum at $\tilde{y} = \pi/b = 3\pi$. For a large value of the amplitude parameter $a = 4/5$ (dotted line), the surface charge density exhibits a local maximum at $\tilde{y} = 3\pi$ and is negative at certain intervals of $\tilde{y}$; in such a case the term “counterion” does not make sense.

Let us continue with the results obtained for the smaller distance $\tilde{d} = \pi/4$. The ratio of dimensionless pressures with the same mean surface charge densities $\bar{P}/\bar{P}_0$ is shown as a function of the amplitude parameter a for two values of the period parameter $b = 2/3$ and $1/3$ in Fig. 4. For $b = 2/3$ the ratio $\bar{P}/\bar{P}_0$ increases monotonically from 1 at $a = 0$ to 1.6766 at the endpoint $a = 0.72$, i.e. modulating surface charge density surprisingly enhances the force between walls. For $b = 1/3$, the ratio $\bar{P}/\bar{P}_0$ decreases very slowly from

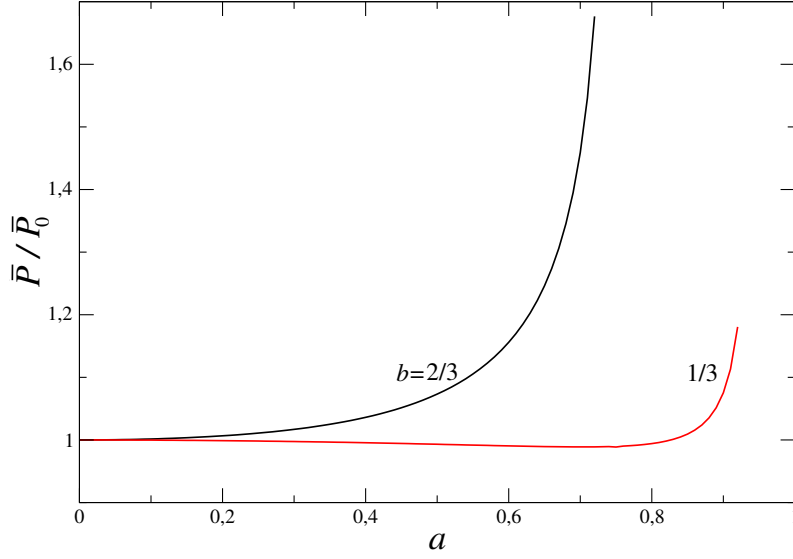


Fig. 4 Dependence of the ratio $\bar{P}/\bar{P}_0$ on the amplitude parameter a for the dimensionless distance between the walls $\tilde{d} = \pi/4$ and the values of the period parameter $b = 2/3, 1/3$.

1 at $a = 0$ to 0.988 at $a = 0.71$ and then increases to 1.181 at the endpoint $a = 0.92$.

For $\tilde{d} = \pi/4$ and $b = 2/3$, the profiles of the normalized modulated surface charge density $\sigma/\langle\sigma\rangle$ as functions of the coordinate $\tilde{y}$ for one period $2\pi/b = 3\pi$ are shown in Fig. 5. For the amplitude parameter $a = 1/5$ (solid line), the surface charge density is always positive and exhibits a minimum at $\tilde{y} = \pi/b = 3\pi/2$. We see that there is an increase in pressure due to modulation of the surface charge density even for the “physical” counterion model with charge distribution $\sigma(\tilde{y}) \geq 0$. For $a = 2/5$ (dashed line), the surface charge density is negative in a short interval around the minimum point $\tilde{y} = 3\pi/2$. For $a = 3/5$ (dotted line), the surface charge density is negative in an interval around the local maximum point $\tilde{y} = 3\pi/2$ and a pair of conjugate minimum points.

The results for a small distance between the walls $\tilde{d} = \pi/8$ are quite similar to the results for $\tilde{d} = \pi/4$. The ratio of dimensionless pressures with the same mean surface charge densities $\bar{P}/\bar{P}_0$ is shown as a function of the amplitude parameter a for two values of the period parameter $b = 2/3$ and $1/3$ in Fig. 6. The ratio is greater than 1 for every allowed value of a .

For $\tilde{d} = \pi/8$ and $b = 2/3$, the normalized modulated surface charge density $\sigma/\langle\sigma\rangle$ as a function of $\tilde{y}$ is always positive for $a = 1/5$ (solid curve) and negative in an interval of $\tilde{y}$ for $a = 2/5$ (dashed curve) and $a = 3/5$ (dotted curve), see Fig. 7.

Explicit analytic results show that for our family of exactly solvable models in the PB limit, surface charge modulation can both increase (small dis-

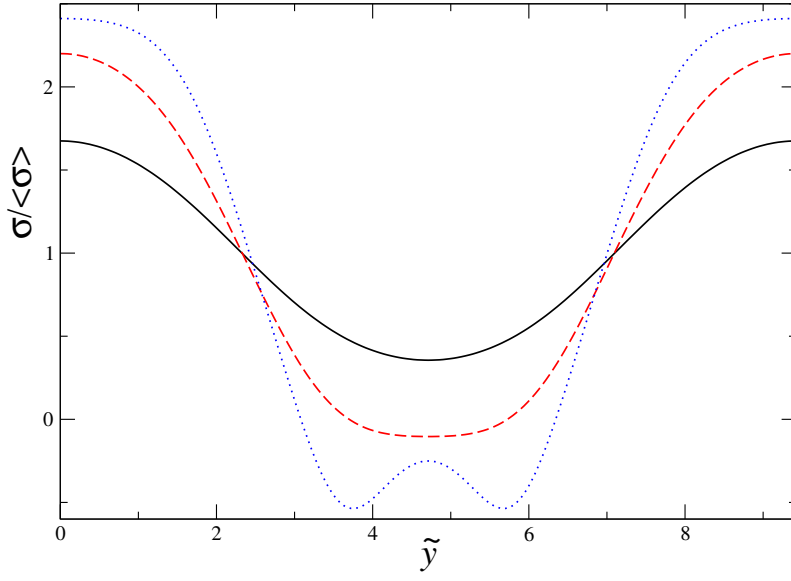


Fig. 5 Plots of the normalized modulated surface charge density $\sigma/\langle\sigma\rangle$ on the dimensionless coordinate $\tilde{y}$ for the dimensionless wall distance $\tilde{d} = \pi/4$ and the period parameter $b = 2/3$, plotted within one period $2\pi/b = 3\pi$. The solid line corresponds to the amplitude parameter $a = 1/5$, the dashed line to $a = 2/5$ and the dotted line to $a = 3/5$.

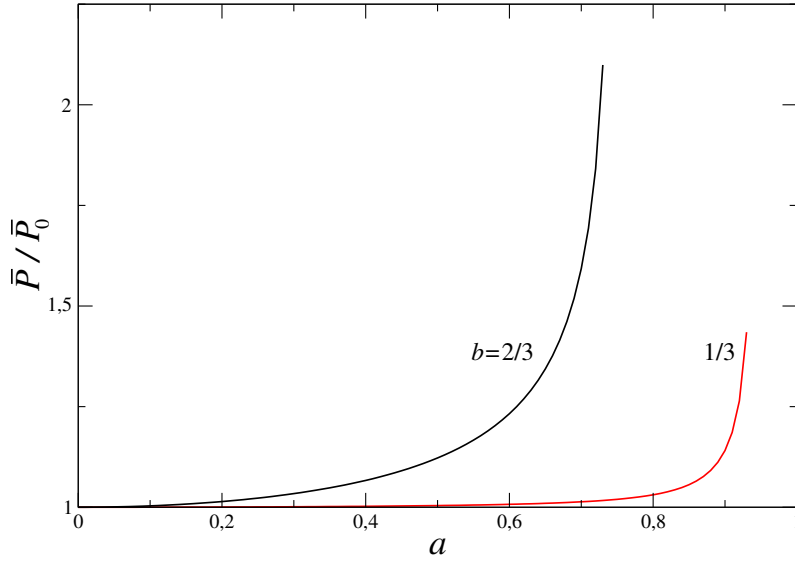


Fig. 6 Dependence of the ratio $\bar{P}/\bar{P}_0$ on the amplitude parameter a for the dimensionless wall distance $\tilde{d} = \pi/8$ and values of the period parameter $b = 2/3, 1/3$.

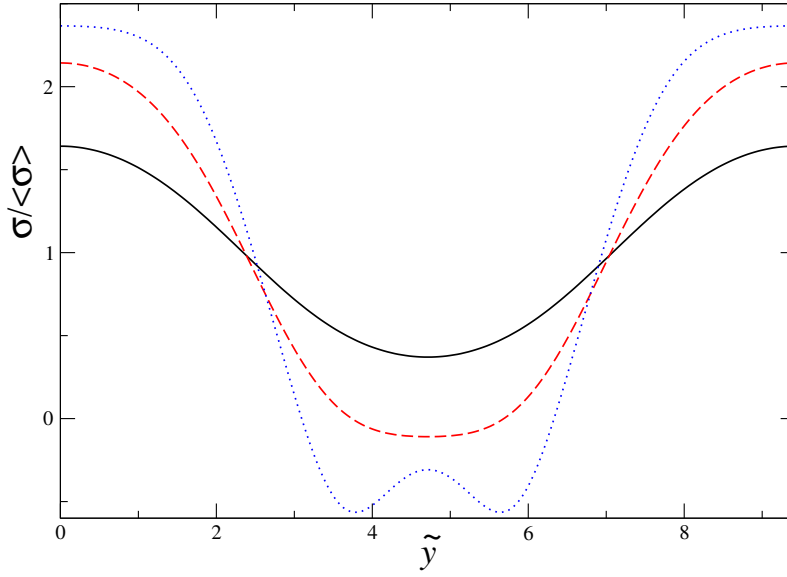


Fig. 7 Plots of $\sigma/\langle\sigma\rangle$ on $\tilde{y}$ for the distance between the walls $\tilde{d} = \pi/8$ and the period parameter $b = 2/3$, plotted within one period $2\pi/b = 3\pi$. The solid line corresponds to the amplitude parameter $a = 1/5$, the dashed line to $a = 2/5$ and the dotted line to $a = 3/5$.

tances) and decrease (large distances) the pressure between two symmetrically charged walls.

8 Conclusion

The exact results are of interest in statistical mechanics because they unambiguously answer important questions without any hypothesis. We tried to answer the question of how the modulation of the surface charge affects the force (or equivalently the pressure) between two symmetric walls at distance d . We restricted ourselves to the mean-field PB theory valid in the high temperature region. Previous approaches based on a combination of analytical perturbation methods and MC simulations [27–30] suggest that the modulation of the surface charge density causes a decrease in the force between the walls. The argument is that an increase in the particle density near the walls with modulated surface charge causes a decrease in the particle density in the midplane between the walls; since the force (pressure) is proportional to this particle density, it decreases. However, as is shown in Eq. (2.25), the local pressure has an additional positive contribution due to the variation of the potential along the midplane, so increasing or decreasing the pressure is questionable. To obtain an exact solution to the problem, we restrict ourselves to modulating the surface charge density along only one direction. This simplification leads to a 2D Liouville equation for the electric

potential (2.11) with boundary conditions (2.12), for which the most general solution exists [40], see Eqs. (5.1) – (5.4). Appropriate choice of the trial functions $Y_1(z)$, $Y_2(z)$ and the constants c_1 , c_2 and c_3 leads to a relatively complicated result for the potential given by Eqs. (5.9) – (5.11). The surface charge density is generated inversely from the electrostatic potential, so our exact results hold for a limited set of models with a specific variation of the surface charge density with the inter-wall distance d . The exact potential is real and free of singularities under a series of conditions for the dimensionless distance between walls $\tilde{d}$, the amplitude and period parameters a and b of the potential, derived in section 6. In practical calculations in the next section 7, these conditions restrict the allowed values of $\tilde{d}$, a and b only slightly. The results for the ratio of the dimensionless pressure $\bar{P}$ for the model with surface charge modulation and the pressure $\bar{P}_0$ for the model with the uniform surface charge density equal to the mean value of the modulated surface charge density $\langle\sigma\rangle$ given by the relation (7.2) are shown for $\tilde{d} = \pi/2$ in Fig. 2, for $\tilde{d} = \pi/4$ in Fig. 4, and for $\tilde{d} = \pi/8$ in Fig. 6. It is clear from these figures that at larger distances, surface charge modulation reduces the force between the walls, while at smaller distances, modulation enhances this force. This is the most important result of this work. The corresponding plots of the normalized surface charge density pictured in Figs. 3, 5 and 7 show that if the amplitude parameter a is small, the surface charge density is positive, while in certain regions of the dimensionless coordinate $\tilde{y}$ it can be negative if a is large. It is important to note that for potential parameters $a = 1/5$, $b = 2/3$ and dimensionless distances $\tilde{d} = \pi/4, \pi/8$, there is an increase in pressure due to modulation of the surface charge density even for “physical” charge distributions with $\sigma(\tilde{y}) \geq 0$.

From the perspective of further possibilities of exploring the present formalism in the future, it seems interesting to look for exact solutions of the 2D Liouville equation that lead to non-symmetric cases of surface charge distribution on walls. This would allow solving, for example, the problem of the impact of a phase shift between modulated surface charges on the force between the walls. Another possibility is to find solutions of the 2D Liouville equation that generate the surface charge as a sum of Dirac delta functions.

In a previous work [39], which dealt with a single wall with surface charge density modulation, exact soliton solutions of the 2D sine-Gordon equation, which represents the PB theory of the Coulomb plasma of $\pm$ charges, were found. It would be interesting to derive analogous soliton solutions for the two-component plasma between two parallel walls.

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Declarations

- Funding Not applicable
- Conflict of interest/Competing interests Not applicable

- Ethics approval and consent to participate Not applicable
- Consent for publication Not applicable
- Data availability Data are available upon a request.
- Materials availability Not applicable
- Code availability Not applicable
- Author contribution Not applicable

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